Lizhong Hu

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Profile

Software Engineering undergraduate in a Sino-American 4+0 dual-degree program (expected graduation: 2027). Experience in backend development, information retrieval and verifiable RAG, Linux/Docker operations, contest problem setting/testing and training coordination, and on-site sports broadcast support.

Education

Quanzhou University of Information Engineering *2023–2027 (expected)*
Software Engineering (Sino-foreign Cooperation) GPA: 3.10/4.00

Slippery Rock University
Information Systems, BS (in progress; expected 2027) GPA: 3.583/4.000
Sino-American 4+0 dual-degree program conducted in China; approved by the Ministry of Education of China.

Research Interests

Primary interests: information retrieval and verifiable RAG. Related interests: algorithms and data structures; software and distributed systems; reliable intelligent information systems.

Projects

NoteLLM: Personal Learning QA with Verifiable RAG Ongoing
Undergraduate thesis

- **Purpose:** Document-based learning QA with verifiable citations and stored source excerpts.
- **Implementation:** Built cosine-similarity retrieval with FastAPI, async SQLAlchemy, PostgreSQL, and pgvector; implemented page-aware extraction and overlapping chunks for PDF, Markdown, and TXT.
- **Citation validation:** Validated citations against retrieved chunks on the server and stored source excerpts for source verification.
- **Evaluation:** A single baseline on 7 synthetic documents and 34 fixed questions recorded 100% source-level Recall@5, 97.1% automatic citation-to-source match, and 339 ms mean retrieval latency.

Evaluation scope: Recall@5 checks whether the expected source appears among the top five retrieved chunks; citation-to-source match checks whether an answer includes at least one validated citation from the expected source. Results measure retrieval and citation-source matching on this synthetic set, not answer accuracy or general-domain performance.
Source code | Evaluation report

ETIPlus: Community Governance & WeChat Management Backend Mar–Apr 2025

- Built a Flask backend with PostgreSQL for landlord credit scoring, rankings, and anomaly audits, and MongoDB for WeChat message storage.
- Developed APIs for group moderation, regex nickname checks, and Excel/CSV imports; automated regression checks with Postman.

DOMjudge Online Judge Deployment & Operations Sep 2024–Present

- Maintain a Linux/Docker-based DOMjudge environment for campus training and programming contests.

Privacy-Oriented Personal Network Management System Personal infrastructure project

- Built a Cloudflare Workers service for Clash/V2Ray profile distribution, ISP ASN-based node selection, configuration caching, and CDN speed testing.
- Automated Linux VPS egress configuration in Python, using WARP IPv6 alongside native IPv4; added backups, repeatable deployment, connectivity checks, and routing-conflict regression tests.

Technical Experience

Sinotech Xinrui (Beijing) Co., Ltd. 7–22 Sep 2026

TOC Operator | Gifu, Japan | 16-day on-site assignment

- Supported football broadcasts at Gifu Nagaragawa Stadium, Japan, for the 20th Asian Games Aichi-Nagoya 2026; host broadcaster: China Media Group-International Media Port (CMG-IMP).
- Checked equipment power, intercom quality, and primary/backup broadcast feeds before matches; coordinated signal verification with the broadcast team and outside broadcast van.
- Monitored TOC signals during matches and provided on-site technical support throughout the 16-day football broadcast assignment.
- Helped engineers isolate a monitor-feed fault to an optical transmitter through receiver replacement and attenuation tests; the feed was restored after post-match replacement and end-to-end verification.

Matiji NOI-Pre Problem Bank | Volunteer Problem Tester Jul–Oct 2024

- Tested programming problems for Matiji's NOI-Pre bank, a collaboration with China Computer Federation (CCF); created boundary cases and checked solution correctness through targeted submissions.
- Investigated and reported test-data and judging defects to support problem quality checks.

Member showcase

Leadership & Activities

ACM Algorithm Club | Organizer Sep 2024–Sep 2025

- Managed day-to-day club operations and helped organize the coastal alliance joint contest, involving seven partner universities.
- ISCPC-AC Cup collegiate programming contest: principal organizer and problem setter.
- 2025 QUIE-Software School ACM Algorithm Contest (2nd Aipin Huiying Science & Technology Festival): one of the organizers and problem setters.

ISCPC-AC Cup | Coastal alliance joint contest

School of International Education | Deputy Director, Learning Department Sep 2024–Sep 2025

- Coordinated faculty lectures, competition events, and student participation.

University ICPC/CCPC Training Team | Coordinator Sep 2024–Sep 2026

- Coordinated university ICPC/CCPC team recruitment, training, selection, and participation logistics.

Supported university team participation in the following eight contests, grouped by year:

- **2026:** The 2026 ICPC China Nanchang National Invitational Programming Contest; 2026 National Invitational of CCPC (Fujian), The 13th Fujian Collegiate Programming Contest; 2026 CCPC Guizhou Provincial Contest and National Invitational.
- **2025:** 2025 CCPC National Invitational (Nanchang) and the 2nd Jiangxi Provincial Collegiate Programming Contest; 2025 ICPC Asia EC Online Preliminary Contest; 2025 CCPC Online Contest; The 2025 ICPC China Guizhou Provincial Programming Contest.
- **2024:** The 2024 ICPC Asia Kunming Regional Contest.

6th College Algorithm Design & Programming Challenge (Spring) Spring 2025

One of the problem setters

- As one of the problem setters, contributed to problem design, solution validation, difficulty calibration, and editorial review.

Competitive Programming & Awards

- 2026 National Invitational of CCPC (Fujian), 13th Fujian Collegiate Programming Contest: First to Solve Award (Problem A; problem-level award).
- The 2024 ICPC Asia Kunming Regional Contest: Honorable Medal; represented Quanzhou University of Information Engineering.
- 15th Lanqiao Cup: First Prize, Fujian Provincial Round, Python Programming (University Group B), 2024.

Technical Skills

Programming & algorithms: C++, Python; algorithms and data structures, competitive programming.

Backend & data: FastAPI, Flask, async SQLAlchemy, PostgreSQL/pgvector, MongoDB; REST API design and integration.

Systems & operations: Linux, Docker, Git, SQL tooling, DOMjudge deployment, Cloudflare Workers/CDN/DNS, WARP egress routing.

Retrieval & AI systems: RAG, LLM API integration, prompt design.

Contest tools: Familiar with Polygon (Codeforces) and Nowcoder; problem preparation, test-data management, and solution verification.

Additional technologies: Java, JavaScript, Go, RuoYi, MySQL.

Development tools & validation: Codex, Claude Code, Antigravity CLI; configure AGENTS.md / CLAUDE.md and shared Skills, decompose tasks, and coordinate subagent review. Validate changes with pytest, Ruff/type checks, Docker Compose, browser checks, and builds.

Contest Archive & Public Records

2025 QUIE-Software School ACM Algorithm Contest — 2nd Aipin Huiying Science & Technology Festival 2025

- Reference problem booklet; personal responsibilities are described under Leadership & Activities.

ISCPC-AC Cup & Coastal Alliance Joint Collegiate Programming Contest

- Contest reference materials; responsibilities for the ISCPC-AC Cup and joint contest are described under Leadership & Activities.

Codeforces Public Profile & Contest Dashboard

- Public participation records and a personal contest dashboard based on public data.

Codeforces profile